Section 3.2: Measurable functions

Section 3.5: Random variables

Section 3.6: The law of a random variable

Section 4.2: The Lebesgue integral (nonnegative functions)

From this section, only the definition of the integral and the property of monotone convergence are discussed; these are needed in the subsequent sections.

Section 4.3: The Lebesgue integral and its properties

Section 4.4: Expectation

Section 3.2: Measurable functions

If we have two measurable spaces $(S_1, \mathcal{S}_1)$ and $(S_2, \mathcal{S}_2)$, a function $f : S_1 \to S_2$ is measurable if and only if

$$f^{-1}(B) \in \mathcal{S}_1 \qquad \text{for every } B \in \mathcal{S}_2.$$

That is, every set we choose from $\mathcal{S}_2$ must be the image of a set from $\mathcal{S}_1$. In other words, every measurable set has a measurable preimage. We say that f is $\mathcal{S}_1/\mathcal{S}_2$-measurable.

Proposition. The function $\mathbf{1}_A$ is measurable for a measurable set A.

Take the measurable space $(S, \mathcal{S})$ and $A \in \mathcal{S}$. This is our space.

For any subset B of the real numbers, one of the following four cases holds:

$$\begin{aligned}
&\text{i)} && 0, 1 \notin B && \longrightarrow && f^{-1}(B) = \mathbf{1}_A^{-1}(B) = \varnothing, \\
&\text{ii)} && 0 \in B,\ 1 \notin B && \longrightarrow && f^{-1}(B) = \mathbf{1}_A^{-1}(B) = A^c, \\
&\text{iii)} && 0 \notin B,\ 1 \in B && \longrightarrow && f^{-1}(B) = \mathbf{1}_A^{-1}(B) = A, \\
&\text{iv)} && 0, 1 \in B && \longrightarrow && f^{-1}(B) = \mathbf{1}_A^{-1}(B) = S.
\end{aligned}$$

In the first case, the preimage is empty. In the fourth case, the preimage contains both the members of A and the members of A^c.

Since $A \in \mathcal{S}$, we see that $\mathbf{1}_A^{-1}(B) \in \mathcal{S}$ for every $B \in \mathcal{B}$. Thus $\mathbf{1}_A$ is measurable.

Introducing generators

Consider a set S. If $\mathcal{C}$ is a family of subsets of S, that is, $\mathcal{C} \subset P(S)$, the σ-algebra generated by $\mathcal{C}$, denoted by $\sigma(\mathcal{C})$, is defined as follows:

$$\sigma(\mathcal{C}) = \bigcap \{\mathcal{A} :\ \mathcal{A} \text{ is a } \sigma\text{-algebra on } S,\ \mathcal{C} \subset \mathcal{A}\} = \mathcal{S}.$$

We say that $\mathcal{C}$ is a **generator** for $\mathcal{S}$.

Thus $\sigma(\mathcal{C})$ is the smallest σ-algebra on S containing $\mathcal{C}$. Every σ-algebra containing $\mathcal{C}$ also contains $\sigma(\mathcal{C})$.

To understand this better, consider the following example.

Example 1. Let $S = \{1, 2, 3\}$ and $\mathcal{C} = \{\{1\}\}$.

We must find the smallest σ-algebra that contains the members of $\mathcal{C}$. We have

$$\varnothing, S \in \mathcal{S}.$$

It also contains $\{1\}$ and $\{1\}^c$. It must be closed under unions. Therefore,

$$\sigma(\{\{1\}\}) = \mathcal{S} = \{\varnothing, \{1\}, \{2, 3\}, S\}.$$

We see that this satisfies all the properties, and hence is the smallest possible σ-algebra with these properties.

Thus $\{\{1\}\}$ is a generator for the space $\mathcal{S}$.

Example 2. The open sets[1] on the real numbers generate the Borel σ-algebra:

$$\sigma(\mathcal{C}) = \sigma\big(\{(-\infty, a) : a \in \mathbb{R}\}\big) = \mathcal{B}(\mathbb{R}).$$

For the proof, it suffices to show that $\sigma(\mathcal{C}) \subset \mathcal{B}(\mathbb{R})$ and $\mathcal{B}(\mathbb{R}) \subset \sigma(\mathcal{C})$.

Proposition: A sufficient condition for measurability of functions

Let $\mathcal{C}$ be a generator of $\mathcal{S}_2$. Then $f : S_1 \to S_2$ is $\mathcal{S}_1/\mathcal{S}_2$-measurable if and only if

$$f^{-1}(B) \in \mathcal{S}_1 \qquad \text{for every } B \in \mathcal{C}.$$

This proposition says that, to show measurability, it is sufficient to check measurability on $\mathcal{C}$, which is much simpler than checking all of $\mathcal{S}_2$.

Proof. One direction is clear from the definition of measurability: if f is $\mathcal{S}_1/\mathcal{S}_2$-measurable, the property in the proposition necessarily holds.

For the converse, in outline, denote by $\mathcal{G}$ the family of sets that satisfy the property, and call these sets "good" sets:

$$\mathcal{G} = \{G \in \mathcal{S}_2 : f^{-1}(G) \in \mathcal{S}_1\}.$$

It is easily checked that $\mathcal{G}$ is a σ-algebra on S_2: it has the properties of a σ-algebra. Also, by assumption, $\mathcal{C} \subset \mathcal{G}$.

Using the definition of a generator and the fact that $\mathcal{G}$ is a σ-algebra containing $\mathcal{C}$,

$$\sigma(\mathcal{C}) \subset \mathcal{G}.$$

But $\mathcal{S}_2 = \sigma(\mathcal{C})$, so $\mathcal{S}_2 \subset \mathcal{G}$. In other words, all of $\mathcal{S}_2$ is "good."

Proposition: Continuous functions are measurable

To prove this, take metric spaces S_1, S_2 with Borel σ-algebras $\mathcal{S}_1, \mathcal{S}_2$. We establish the proposition using Example 2 and the preceding proposition.

[1] An open set is a set such that, for each of its points, a neighborhood of that point can be found that lies entirely inside the set. For example, $[a, b]$ is not open, but (a, b) is open.

Section 3.5: Random variables

A **probability space** is a triple $(\Omega, \mathcal{A}, \mathbb{P})$, where Ω is a set, $\mathcal{A}$ is a σ-algebra on Ω, and $\mathbb{P}$ is a probability measure.

A **random variable** on the probability space $(\Omega, \mathcal{A}, \mathbb{P})$ is a measurable function $X : \Omega \to S$, where $(S, \mathcal{S})$ is a measurable space.

The symbols $\mathbb{P}(X = x)$ and $\mathbb{P}(X \in B)$ are defined by

$$\mathbb{P}(X = x) = \mathbb{P}(X^{-1}\{x\}) = \mathbb{P}(\{\omega \in \Omega : X(\omega) = x\}),$$

$$\mathbb{P}(X \in B) = \mathbb{P}(X^{-1}(B)) = \mathbb{P}(\{\omega \in \Omega : X(\omega) \in B\}).$$

A **random vector** on the probability space $(\Omega, \mathcal{A}, \mathbb{P})$ is a measurable function $X : \Omega \to \mathbb{R}^n$. We usually regard $\mathbb{R}^n$ as a measurable space with the Borel σ-algebra $\mathcal{B}(\mathbb{R}^n)$.

Note. If S is a topological space, so that neighborhoods and hence open sets are defined, we may equip it with the Borel σ-algebra $\mathcal{B}(S)$. Otherwise, other σ-algebras must be used.

Section 3.6: The law of a random variable

If $X : \Omega \to S$ is a random variable defined on a probability space $(\Omega, \mathcal{A}, \mathbb{P})$, and $(S, \mathcal{S})$ is a measurable space, the **law**, or distribution, of X is a function $\mu : \mathcal{S} \to [0, 1]$ defined by

$$\mu(B) = \mathbb{P}(X^{-1}(B)).$$

Intuition. The probability $\mathbb{P}$ is defined on subsets of Ω, which are usually not tangible: they are the input to X. But μ transfers the probabilities to subsets of S, which are usually tangible: they are the output of X.

The importance of this is that we usually work with random variables. When $\mathbb{P}$ assigns a probability to an experiment or a random process, describing the outcomes of that process is very difficult, and we do not even need to describe them. Instead, we call a specific outcome of that experiment an X.

Proposition. The law of X is a probability measure on the measurable space $(S, \mathcal{S})$.

Proof. By the definition,

$$\text{i)} \qquad \mu(\varnothing) = \mu(X^{-1}(\varnothing)) = \mathbb{P}(\varnothing) = 0 \quad \longrightarrow \quad \mu(\varnothing) = 0.$$

ii) Also, if $B_i \in \mathcal{S}$ are disjoint sets, then for $i \neq j$,

$$X^{-1}(B_i) \cap X^{-1}(B_j) \overset{*}{=} X^{-1}(B_i \cap B_j) = X^{-1}(\varnothing) = \varnothing.$$

We do not prove the equality marked $*$ here, but it is readily understood from the definition of a preimage and is intuitively clear.

Thus the preimages $X^{-1}(B_i)$ are also pairwise disjoint. Hence,

$$\begin{aligned}\mu\left(\bigcup_i B_i\right) &= \mathbb{P}\left(X^{-1}\left(\bigcup_i B_i\right)\right) \\ &\overset{*}{=} \mathbb{P}\left(\bigcup_i X^{-1}(B_i)\right) \\ &= \sum_i \mathbb{P}(X^{-1}(B_i)) = \sum_i \mu(B_i).\end{aligned}$$

Thus μ has the property of countable additivity and is consequently a measure.

$$\text{iii)} \qquad \mu(S) = \mathbb{P}\big(X^{-1}(S)\big) = \mathbb{P}(\Omega) = 1.$$

Therefore μ is a probability measure.

The law of X is usually written as $\mu = \mathbb{P} \circ X^{-1}$. It is called the **pushforward** of $\mathbb{P}$ by X: it moves the probability law to the output space.

Proposition. For every probability measure μ on $(S, \mathcal{S})$, there exist a probability space $(\Omega, \mathcal{A}, \mathbb{P})$ and a random variable $X : \Omega \to S$ such that the law of X equals μ.

Proof. Define

$$(\Omega, \mathcal{A}, \mathbb{P}) := (S, \mathcal{S}, \mu),$$
$$X : \Omega \to S, \qquad X(\omega) = \omega.$$

Then, for every $B \in \mathcal{S}$, we have $X^{-1}(B) = B$. Consequently,

$$\underbrace{\widetilde{\mu}(B)}_{\text{law of } X} = \mu(X^{-1}(B)) = \mu(B).$$

Thus μ is the law of B.

(We have taken the distribution itself as the initial space and X as the identity function on it.)

Section 4.2

Introducing the integral of a nonnegative function with respect to a measure, and the monotone convergence theorem, without proof

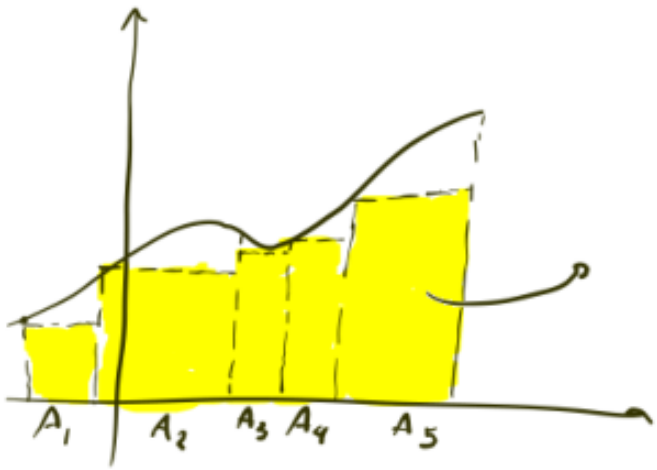
The integral of a nonnegative measurable function $f : S \to \mathbb{R}_+$ with respect to a measure μ on the measurable space $(S, \mathcal{S})$ is defined by

$$\int_S f \, d\mu = \sup_{\{A_i\}} \sum_i \left(\inf_{s \in A_i} f(s) \right) \mu(A_i),$$

where the supremum is over all partitions $\{A_i\}$ of S into finite or countably infinite families of sets.

This integral is called the **Lebesgue integral**.

Intuition. This definition is similar to the Riemann definition. We partition S, then in each part multiply the smallest value of the function in that part, its infimum, by the measure of that part, its area. We do this for every possible partition and take the largest value, the supremum. Intuitively:



The supremum is taken over all finite and countably infinite partitions.

The integral above is also written in the following forms:

$$\int_S f \, d\mu = \int_S f(s) \, \mu(ds) = \mu(f).$$

Monotone convergence theorem

For nonnegative measurable functions $f, f_n : S \to \mathbb{R}_+$,

$$f_n \uparrow f \quad \Rightarrow \quad \mu(f_n) \uparrow \mu(f).$$

We do not prove the theorem here. However, from the figure above, it can be understood completely intuitively. Note that f_n and f are nonnegative.

Section 4.3: Extending the definition of the integral to all functions

The integral of a measurable function $f : S \to \mathbb{R}$ is defined by

$$\int_S f \, d\mu = \int_S f_+ \, d\mu - \int_S f_- \, d\mu,$$

where the positive and negative parts of f are

$$f_+ = \max\{f, 0\}, \qquad f_- = \max\{-f, 0\}.$$

Both f_+ and f_- are nonnegative, so we can use the definition from the previous section for them. Intuitively, the definition is the same as before, except that for the negative parts we consider the negated function and subtract the result.

We state the properties of the integral briefly and without proof.

1. Monotonicity:

$$f \leq g \quad \Rightarrow \quad \mu(f) \leq \mu(g).$$

2. Integrability:

$$f \text{ is integrable} \quad \iff \quad \int_S |f| \, d\mu < \infty.$$

3. Linearity:

$$\mu(af + bg) = a\mu(f) + b\mu(g),$$

where $a, b \in \mathbb{R}$ and f, g belong to a real vector space.

4. Triangle inequality:

$$|\mu(f)| \leq \mu(|f|).$$

Dominated convergence theorem

For real-valued measurable functions $f_n, f : S \to \mathbb{R}$, with $f_n < g$ and $\int_S |g| \, d\mu < \infty$,

$$f_n \to f \quad \Rightarrow \quad \int_S f_n \, d\mu \to \int_S f \, d\mu.$$

Intuitively, we can say that if $f_n \to f$, then its integral with respect to the measure μ will also equal the integral of f with respect to μ.

A comment on integration in the Lebesgue sense

The Lebesgue integral extends the concept of integration to more general functions. In brief, the difference between the Riemann and Lebesgue integrals is that the Riemann integral is based on the length of intervals, whereas the Lebesgue integral is based on the measure of sets. Consequently, there are functions that are not Riemann integrable but whose Lebesgue integrals exist.

For example, consider the function $f : [0,1] \to \{0,1\}$ defined by

$$f(x) = \begin{cases} 1, & x \notin \mathbb{Q}, \\ 0, & x \in \mathbb{Q}. \end{cases}$$

Clearly, this function is not Riemann integrable: whichever interval Δx we choose, f does not take one specific value throughout it, and a Riemann sum cannot be calculated for it.

In the Lebesgue sense, however, the measure of the rational numbers is 0, and the measure of their complement is the Lebesgue measure, equal to the length of the interval, namely one. Thus the function is Lebesgue integrable, and its integral is

$$1 \times \mu([0,1]) = 1.$$

Section 4.4: Expectation

The expectation of a random variable $X : \Omega \to \mathbb{R}$ on a probability space $(\Omega, \mathcal{A}, \mathbb{P})$ is defined by

$$\boxed{\mathbb{E}X = \int_\Omega X \, d\mathbb{P}}.$$

A random variable is integrable if $\mathbb{E}|X| < \infty$.

Properties of expectation

These properties follow from the corresponding properties of the integral.

1. **Linearity:**
$$\mathbb{E}(aX + bY) = a\mathbb{E}(X) + b\mathbb{E}(Y).$$
2. **Monotonicity:**
$$X \leq Y \quad \Rightarrow \quad \mathbb{E}X \leq \mathbb{E}Y.$$
3. **Dominated convergence:** If $\mathbb{E}|X_n| < \infty$,
$$X_n \to X \quad \Rightarrow \quad \mathbb{E}X_n \to \mathbb{E}X.$$
4. **Bounded convergence (a case of the preceding item):** If $|X_n| < c$,
$$X_n \to X \quad \Rightarrow \quad \mathbb{E}X_n \to \mathbb{E}X.$$

And if $X : \Omega \to \mathbb{R}_+$ is nonnegative:

Monotone convergence:
$$X_n \uparrow X \quad \Rightarrow \quad \mathbb{E}X_n \uparrow \mathbb{E}X.$$